An established traffic backbone for an elementary VSL task

Literature comparison, necessary adaptation and methodological findings

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1. What a suitable backbone must provide

Can the intended information and gradient questions be studied using an established variable speed limit (VSL) model with very few changes? The motivation is to keep the traffic-model justification proportionate to an elementary verification of a computational method. A backbone should supply the traffic state and most evolution laws; simplicity concerns the assumptions and coupled rules we must author and justify, rather than the number of references cited.

Consider a road with an ending lane E, its adjacent continuing lane A and a farther continuing lane B. Two initial states can have identical E traffic and identical combined A+B traffic at every road position, but different A/B splits. We ask whether that concealed split changes merging service, whether knowing it can improve a common speed decision, and whether complete travel-time sensitivity can guide that decision. These are separate claims. The objective is total system time, counting every initial unit until exit. A suitable task must conserve traffic and respect shared receiving space. Neither a plausible lane effect nor a differentiable formula establishes a control advantage.

2. What the closest published models offer

The inspected literature offers two substantive routes: use an aggregate VSL setting and ask about traffic location or history, or retain lane densities to ask specifically about A/B loading. Table 1 compares what each route already supplies and what our question would require.

Table 1. Published starting points and their fit to the question

Published model	What is already supplied	Direct finding for this review
Jin and Jin [1]	Spatial aggregate traffic and a link-average approximation; VSL entry control, threshold capacity drop and upstream holding queues.	The simplest inspected route if the hidden information is *where* traffic lies along the road. It has no A/B lane split; its discharge switch still needs sensitivity qualification.
Wei, Han and Wang [2]	Link traffic with entry/speed history, interacting speed cohorts and capacity drop; advice acts on entering traffic.	A serious history-aware aggregate alternative. Its native state omits A/B loading, and its action timing differs from common advice throughout our road.
Nagalur Subraveti et al. [3]	Densities in successive lane cells, lane-change incentives and a lane-drop application; congestion dynamics draw on Han et al. [4].	Best fit among the inspected lane backbones for retaining the A/B question. Add the VSL sending restriction [5] and explicitly qualify movement realization; direct source transcription is insufficient.
Carey et al. [6]	Multilane cell transmission, intended lane changes, mandatory movements and receiving-allocation alternatives.	Useful support for conservative receiving and fallback rules. As the main backbone, more of the VSL and permanent lane-end task would need specifying.

Jin and Jin therefore offer a genuine simplification, with a different information contrast: equal total traffic can be distributed differently along a road. A fair information study would use the same spatial traffic model with different observations. Comparing their spatial and link-average simulators would also change the dynamics. Likewise, lawful inference from aggregate history must be admitted; aggregate models should not be reduced to an arbitrary equal-lane guess.

For the chosen lane-loading question, the Subraveti framework supplies both the missing state and reasons to redistribute it, including approaching the lane end, density differences, keep-right behavior and cooperation. Its Han lineage supplies a compatible starting point for congestion and VSL budgets [4], [5]. This is a bounded comparison of inspected methods, not a proof that no simpler native model exists. The remaining issue is whether the lane framework can be completed with a limited, explicit change.

3. How the selected model is completed

Concrete task and assumptions. The elementary member has ten 100 m road sections, six-second updates and a permanent three-to-two lane drop. A finite cohort is initialized inside the road, independently of one constant common advice of 30–60 km/h. There are no later arrivals; crossing the common free A/B outlet line completes a trip. E, A and B denote current lane stocks: early E-to-A traffic becomes ordinary A traffic. The root assumptions are:

- Continuous mass, fixed geometry and a common lane flow–density law; the time step respects sending and backward-wave storage bounds.
- Local movements use pre-update traffic and space, advance one section, and are applied simultaneously. New arrivals or freed space are not reused within the same update.
- Equal positive nominal priorities and unit receiving-space consumption per unit of traffic; merging has no additional headway multiplier.

Retained and adapted ingredients.

Lane-density preferences retain the Subraveti framework [3]; its same-author thesis clarifies density lookahead [7]. Full Han budgets bound available traffic and receiving service, with advice capping sending [4], [5]. Thus advice influences later merge service through evolving traffic states. It does not directly prescribe a merge fraction or change nominal priorities. Finite-domain, empty-lane and traffic-mode conventions are specified explicitly for this member.

The substantive replacement is movement realization. Review hand cases exposed donor/receiver budget conflicts in direct transcription of the displayed native coupling; these are not claims about recovered author code. The selected composition first normalizes all a lane's requests against its one sending allowance. Requests into a destination then share its one receiving allowance: small requests are satisfied, with unused entitlement redistributed. Remaining traffic gets one opportunity to continue in its own lane. At E's end there is no forward escape: it enters A's outlet or waits. This adapts Carey receiving/fallback principles and Wright priority sharing [6], [8]. The complete composition is ours; changing lane also advances traffic into the next road section.

Every realized transfer spends one donor budget and one receiver budget. Subtracting and adding that same mass preserves conservation, while the old-space bound protects storage. These joins make a complete model without adding another physical mechanism. They do not imply maximum throughput, equivalent congestion waves or inherited calibration. Calling this one replacement component does not remove its assumptions. The closed-cohort, one-advice comparison is also our task design, not an unchanged published benchmark. The companion formulation retains the complete equations and hand cases.

4. Recommendation and what the evidence establishes

Complete accounting and qualified sensitivity are established analytically. Under these assumptions, positive service and free outlets eventually bring the cohort into an unconstrained regime. Remaining mass need not become exactly zero in finite time: its distance to the exit gives an exact remaining total system time. Consequently the objective counts the full trip rather than discarding a small residual. For the fixed finite model, the complete objective has smooth rational formulas between finitely many possible advice switches. Its derivative includes earlier traffic changes, residual-state changes and the direct residual travel-time effect. At switches, jumps or unequal slopes may remain; derivatives can also be zero. This is a fixed-step analytical result, not numerical gradient verification.

Lane information changes service; its control benefit is unproved. The reviewed A/B pair has the same E and A+B profiles but different actual merge service and exit flow. Advice is chosen before that informative discharge; no identical hidden prehistory is asserted. The reduced observation loses predictive information under this model. One extra A density in the last road section identifies that pair completely, so it proves no need for all lane fields or discrete order. Nor does different service establish different optimal advice. In the sufficient small-cohort regime where all movements remain unconstrained, equal traffic at each road position gives identical full costs: maximum advice is optimal and lane information has zero action value. The congested diagnostic lies outside that regime; its action gap is open.

The resulting recommendation is conditional. Use the published lane framework, its congestion/VSL lineage and the explicit movement replacement as a candidate for the selected information question. Keep a model-aware E+(A+B) observer, the sufficient one-scalar augmentation, maximum advice and equal-information scalar search as strong comparisons. They separate information value from optimizer value. The thesis omits bounded lane-change acceleration [7]; no earlier slow-entry/recovery explanation or positive-action certificate transfers. Numerical correctness, physical fidelity and useful-gradient performance remain future evidence. The review establishes a coherent route with visible adaptation costs; it does not establish that this route is globally simplest or best.

References

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